

Probability (Set A) — MCQ Worksheet

Mathematics • Chapter 15 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Probability of an event E is given by:

- | | |
|--|------------------------|
| (A) Number of favourable outcomes / Total outcomes | (B) Total / favourable |
| (C) Favourable $\times$ Total | (D) Favourable + Total |

Q2. Probability of any event E satisfies:

- | | |
|----------------|--------------------------|
| (A) $P(E) > 1$ | (B) $0 \leq P(E) \leq 1$ |
| (C) $P(E) < 0$ | (D) $P(E) = 2$ |

Q3. Probability of a sure event is:

- | | |
|-------|----------------|
| (A) 0 | (B) 1 |
| (C) 2 | (D) Cannot say |

Q4. Probability of an impossible event is:

- | | |
|--------------|---------|
| (A) 0 | (B) 1 |
| (C) ∞ | (D) 0.5 |

Q5. The sum of probabilities of all elementary events of an experiment is:

- | | |
|-------|-------------------|
| (A) 0 | (B) $\frac{1}{2}$ |
| (C) 1 | (D) 2 |

Q6. $P(E) + P(\text{not } E) =$

- | | |
|-------|-------------------|
| (A) 0 | (B) $\frac{1}{2}$ |
| (C) 1 | (D) 2 |

Q7. If a die is rolled, probability of getting an even number is:

- | | |
|-------------------|-------------------|
| (A) $\frac{1}{2}$ | (B) $\frac{1}{3}$ |
| (C) $\frac{1}{6}$ | (D) $\frac{2}{3}$ |

Q8. If a die is rolled, probability of getting a number greater than 4 is:

- | | |
|-------------------|-------------------|
| (A) $\frac{1}{2}$ | (B) $\frac{1}{3}$ |
| (C) $\frac{1}{6}$ | (D) $\frac{2}{3}$ |

Q9. If a die is rolled, probability of getting a 7 is:

- | | |
|-------------------|-------------------|
| (A) $\frac{1}{6}$ | (B) 0 |
| (C) 1 | (D) $\frac{1}{7}$ |

Q10. A coin is tossed twice. Probability of getting at least one head is:

- | | |
|-------------------|-------------------|
| (A) $\frac{1}{4}$ | (B) $\frac{1}{2}$ |
| (C) $\frac{3}{4}$ | (D) 1 |

Q11. A coin is tossed once. $P(\text{head}) =$

- | | |
|-------|-------------------|
| (A) 1 | (B) $\frac{1}{2}$ |
| (C) 0 | (D) 2 |

Q12. Two coins are tossed. Probability of getting two heads is:

- | | |
|-------------------|-------------------|
| (A) $\frac{1}{2}$ | (B) $\frac{1}{3}$ |
| (C) $\frac{1}{4}$ | (D) 1 |

Q13. From a well-shuffled pack of 52 cards, probability of getting a king is:

- | | |
|--------------------|---------------------|
| (A) $\frac{1}{13}$ | (B) $\frac{1}{52}$ |
| (C) $\frac{4}{13}$ | (D) $\frac{12}{52}$ |

Q14. From a well-shuffled pack, probability of getting a red card is:

- | | |
|-------------------|--------------------|
| (A) $\frac{1}{4}$ | (B) $\frac{1}{2}$ |
| (C) $\frac{3}{4}$ | (D) $\frac{1}{13}$ |

Q15. From a pack of cards, probability of getting a face card is:

- (A) $\frac{1}{13}$
(C) $\frac{4}{13}$

- (B) $\frac{3}{13}$
(D) $\frac{12}{13}$

Q16. From a pack of cards, probability of getting an ace is:

- (A) $\frac{1}{13}$
(C) $\frac{4}{13}$

- (B) $\frac{1}{52}$
(D) $\frac{1}{4}$

Q17. A box contains 5 red, 4 green, 3 blue balls. Probability of drawing a red is:

- (A) $\frac{5}{12}$
(C) $\frac{3}{12}$

- (B) $\frac{4}{12}$
(D) $\frac{1}{3}$

Q18. A bag contains 3 white and 4 red balls. Probability of drawing a white ball is:

- (A) $\frac{3}{7}$
(C) $\frac{1}{2}$

- (B) $\frac{4}{7}$
(D) $\frac{2}{5}$

Q19. In a class of 50 students, 30 like cricket. Probability that a randomly chosen student likes cricket is:

- (A) $\frac{3}{5}$
(C) $\frac{1}{5}$

- (B) $\frac{2}{5}$
(D) $\frac{1}{2}$

Q20. Two dice are thrown. Probability that sum = 7 is:

- (A) $\frac{1}{12}$
(C) $\frac{1}{9}$

- (B) $\frac{1}{6}$
(D) $\frac{1}{3}$

Q21. Two dice are thrown. Probability that both show the same number is:

- (A) $\frac{1}{6}$
(C) $\frac{1}{12}$

- (B) $\frac{1}{3}$
(D) $\frac{6}{36} = \frac{1}{6}$

Q22. From digits 1, 2, 3, 4, 5, two-digit numbers are formed. Probability that the number is even is:

- (A) $\frac{2}{5}$
(C) $\frac{1}{2}$

- (B) $\frac{2}{4}$
(D) $\frac{3}{5}$

Q23. A bag contains 3 red and 2 black balls. $P(\text{black}) =$

- (A) $\frac{2}{5}$
(C) $\frac{1}{5}$

- (B) $\frac{3}{5}$
(D) $\frac{4}{5}$

Q24. Probability of getting a prime number when a die is thrown is:

- (A) $\frac{1}{3}$
(C) $\frac{2}{3}$

- (B) $\frac{1}{2}$
(D) $\frac{1}{6}$

Q25. If the probability of a defective bulb is 0.05 in a lot of 200, the expected number of defectives is:

- (A) 5
(C) 15

- (B) 10
(D) 20